COMP2611 COMPUTER ORGANIZATION COURSE INFORMATION (FALL 24)

Teaching Team

- **A team of 8 (instructor, instruction assistant and postgraduate teaching assistants)**
- **Our promise: always try our best to support high-quality teaching and learning**

Prof. Cindy Li
Instructor L1 & L2

Mr. Vinayak KHURANA
Instructional Assistant

- 3 UGTAs as lab helpers and provide weekly Q&A sessions

Course Websites and Schedule

- **Course webpage https://course.cse.ust.hk/comp2611/**
  - All course learning materials, visit it regularly for updates
- **Course Canvas https://canvas.ust.hk/courses/58141**
  - Homework/PA submission, grade checking
- **Course Piazza https://piazza.com/class/m0dhxme0l8e3vh**
  - Online Q&A forum, learning community

- **2 x 1.5-hour F2F lectures**
- **1-hour F2F tutorial**
- **1-hour F2F lab**

- **Tutorial and Lab start in week 3 (week of Sep 16 – Sep 20)**
- **Tutorials and labs are useful supplement and nice revision of the lectures**

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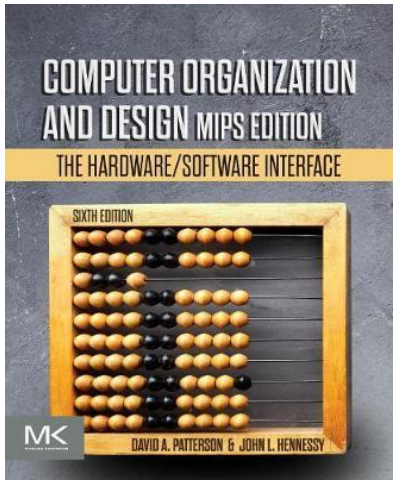
Textbook and Grading Scheme



- **Computer Organization and Design MIPS Edition**
  - 6th edition, ISBN 9780128201091
- **Grading scheme**
  - **4 individual written homework 15% (15%)**
    - Released in week 4, week 6, week 8, and week 11, normally due in one or two weeks
  - **Midterm exam 30%**
    - Oct 25 Fri evening, venue TBD, week 8
  - **1 individual programming project 15%**
    - Released in week ~9, due in one month (deadline: last day of Fall)
  - **Final Exam 40%**
    - mid/late Dec

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Course Schedule Overview

September

1	1	2	3	4	5	6	7
2	8	9	10	11	12	13	14
3	15	16	17	18	19	20	21
4	22	23	24	25	26	27	28
5	29	30					

October

5			1	2	3	4	5
6	6	7	8	9	10	11	12
7	13	14	15	16	17	18	19
8	20	21	22	23	24	25	26
9	27	28	29	30	31		

November

9						1	2
10	3	4	5	6	7	8	9
11	10	11	12	13	14	15	16
12	17	18	19	20	21	22	23
13	24	25	26	27	28	29	30

- Tutorial and Lab start in week 3
  - T1 on Mon 17:00 – 17:50, T2 on Tue 18:00 – 18:50
  - LA1 on Fri 12:30 – 1:20, LA2 on Wed 11:00 – 11:50
- Week 4: Homework 1 release
- Week 6: Homework 2 release
- Week 8: Homework 3 release
- Week 8: Midterm (Oct 25 Evening)
- Week 9: PA release
- Week 11: Homework 4 release

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Academic Integrity

- Course homework and project should be individual work; both ‘provider’ and ‘copier’ will be penalized equally and harshly
- University guideline
- https://registry.hkust.edu.hk/resource-library/academic-integrity

- If you encounter difficulties in learning, do approach us. We can arrange UGTAs to follow up.

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Questions left Unanswered

- **In COMP2011 (or similar high-level programming courses), you've learned**
  - Develop a program to solve problem

- **But questions remain as**
  - How does a computer work inside?
  - How can we get electricity to perform abstract tasks like adding and storing numbers?
  - What happens when our programs are executed by a computer?
  - How do we design programming languages and hardware to work together?
  - How do we make computers faster?

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Course Learning Outcomes

- **Upon completion of the course, students are expected to be able to:**
  - Understand the basic concepts of digital logic and build the small circuits involved in computer systems
  - Describe the interaction between software and hardware and instruction set architecture (ISA)
  - Write and execute small programs of a few hundred lines in assembly language
  - Define the basic concepts of modern computer hardware, including processor(datapath, control) and memory
  - Describe the organizational paradigms that determine the capability and performance of computer systems

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Topics to be covered

- **Brief Digital Circuit**
  - Topic 1: Digital Logic (combinational and sequential)
- **Data Representation and Arithmetic**
  - Topic 2: Data Representation (integer, fractional, character)
  - Topic 3: Computer Arithmetic (addition, subtraction, multiplication, division) and ALU
- **Instruction Set Architecture and Assembly**
  - Topic 4: Instruction Set Architecture (ISA) and MIPS Assembly Language
- **Computer Architecture**
  - Topic 5: Processor (datapath and control) and pipeline
  - Topic 6: Memory System (cache, virtual memory)

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